

The Thermodynamic Identity Governing the Virial Theorem: Derivation from First Principles and Evidence Across 37 Orders of Magnitude

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March 2026

Zenodo DOI: [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042)

Abstract

The virial theorem, established by Clausius in 1870, states that for any stable bound system under a $1/r$ potential, the time-averaged kinetic and potential energies satisfy $2\langle K \rangle + \langle V \rangle = 0$. The theorem has been verified across 37 orders of magnitude in physical scale, but one question has persisted: what is $\langle K \rangle$? The virial theorem identifies $\langle K \rangle$ as the average kinetic energy of bound particles. It does not identify what physical process produces the $1/2$ partition, nor what happens thermodynamically to the energy released during the transition from the unbound to the bound state.

We show that the virial theorem and Landauer's principle are two expressions of the same underlying thermodynamic identity. The first law, the second law, Landauer's principle, and the equilibrium condition of the $1/r$ potential together require:

$$E_{\text{Landauer}} = Q = -E_f = \langle K \rangle,$$

where Q is the heat released irreversibly at the virialization boundary, ΔS is the entropy produced, and E_f is the total energy of the bound state. This identifies $\langle K \rangle$ for the first time as a thermodynamic quantity: the Landauer cost of the system's transition from the unbound to the bound state. The thermodynamic identity is:

$$\langle K \rangle = Q = T \Delta S = E_{\text{Landauer}}. \quad (1)$$

The virial theorem $2\langle K \rangle + \langle V \rangle = 0$ is the mechanical expression of this identity. The identity and the virial theorem are two descriptions of the same physical condition at the boundary between unbound and bound states.

The data are consistent with this identity across 37 orders of magnitude in physical scale, from hydrogen atoms at 10^{-10} m to the cosmological coupling constant $\beta_m = \Omega_m/2$, for which 17 independent MCMC chains against the full

Planck 2018 likelihood [Planck Collaboration, 2020] return $\beta_m = 0.1583 \pm 0.0033$ against the predicted 0.1577, consistent at 0.2σ with zero free parameters. The Informational Actualization Model (IAM) [Mahaffey, 2026f] is one framework that instantiates this identity at cosmological scales, recovering $\beta_m = \Omega_m/2 = 0.1575$ with no fitted parameters.

Three of the four fundamental forces satisfy this identity within their respective domains. The weak nuclear force is a structural exception, excluded not by coincidence but by physical necessity: the force whose function is to enable irreversible transitions between stable states cannot simultaneously satisfy an equilibrium condition. This exception is itself evidence for the identity’s physical content.

We propose that this thermodynamic identity constitutes a fundamental law of physics of which the virial theorem is the mechanical special case.

Keywords: virial theorem; Landauer’s principle; thermodynamic identity; gravitational decoherence; entropy; bound states; cosmological constant; Informational Actualization Model

1 Introduction

1.1 The unexplained 1/2

The virial theorem has governed the energy partitioning of bound physical systems since Clausius stated it mathematically in 1870 [Clausius, 1870]. For any stable system bound by a $1/r$ potential, the time-averaged kinetic and potential energies satisfy:

$$2\langle K \rangle + \langle V \rangle = 0, \quad \langle K \rangle = \frac{1}{2}|\langle V \rangle|. \quad (2)$$

The theorem is universally verified — from the hydrogen atom (10^{-10} m) to galaxy clusters (10^{23} m) and, as shown below, to the cosmological coupling constant (10^{26} m) — a span of 37 orders of magnitude. It is one of the most broadly confirmed results in physics.

Yet a question has gone unanswered for 154 years: what is K ?

The virial theorem identifies K as the time-averaged kinetic energy of the bound particles. It establishes the ratio $K/|V| = 1/2$ as a mathematical consequence of the $1/r$ potential form via the equations of motion. It does not explain why the energy balance at equilibrium requires exactly this ratio, nor what physical process produces it, nor what happens to the energy that is not retained as kinetic energy during the transition from the unbound to the bound state.

Clausius derived Equation 2 from Newton’s second law. The derivation is mechanical. It describes the equilibrium state. It is silent on the thermodynamic content of the transition that produces the equilibrium state, and silent on the physical identity of the energy released during that transition.

The thermodynamics of gravitational systems has a substantial history. Lynden-Bell and Wood [Lynden-Bell and Wood, 1968] identified the gravothermal catastrophe; Padmanabhan [Padmanabhan, 1990] analyzed statistical mechanics of self-gravitating systems; Saslaw [Saslaw, 1985] developed the statistical thermodynamics of gravitational clustering across stellar and galactic scales. These works study the thermodynamic properties of systems already in or near virial equilibrium. What has not been shown is that the virial theorem and Landauer’s principle are two expressions of the same identity, and that this identity provides the physical meaning of $\langle K \rangle$ as the Landauer cost of the virialization transition itself.

1.2 The present work

This paper shows that the virial theorem is the mechanical expression of a thermodynamic identity, and uses that identity to provide the first physical identification of $\langle K \rangle$. The argument proceeds in four steps:

1. The transition from an unbound to a bound state under a $1/r$ potential releases heat Q irreversibly at the boundary. By the first law, $Q = -E_f$.
2. By the second law and Landauer’s principle [Landauer, 1961], the thermodynamic cost of this irreversible transition is $E_{\text{Landauer}} = T_{\text{vir}} \Delta S = Q$. The virial temperature cancels exactly, making this result scale-invariant.
3. The equilibrium condition of the $1/r$ potential — expressed by the virial theorem itself — requires $-E_f = \langle K \rangle$ at stable bound equilibrium.
4. Combining steps 1–3: $E_{\text{Landauer}} = Q = -E_f = \langle K \rangle$. This identifies $\langle K \rangle$ as the Landauer cost of the virialization transition. The virial theorem is the mechanical statement of this thermodynamic identity.

The result is that K is not merely a mechanical quantity. It is the thermodynamic cost of the system’s transition from the unbound to the bound state — the heat released irreversibly and encoded permanently as entropy at the virialization boundary. The virial theorem is the mechanical expression of this thermodynamic identity.

Section 2 presents the derivation. Section 3 states the identity formally and compares it to the virial theorem. Section 4 presents the evidence across 37 orders of magnitude. Section 5 analyzes the four fundamental forces. Section 6 presents the MCMC confirmation as the strongest quantitative test. Section 7 discusses implications.

2 Derivation

2.1 Setup

Consider a system of total mass M transitioning from an unbound state to a stable bound equilibrium state under a $1/r$ gravitational potential. We adopt the natural reference state for a $1/r$ potential: the unbound state at rest at infinity, with total energy $E_i = 0$.

The bound equilibrium state has total energy:

$$E_f = \langle K \rangle + \langle V \rangle. \quad (3)$$

No assumption about the ratio $\langle K \rangle / |\langle V \rangle|$ is made at this stage. The thermodynamic argument will identify the physical content of that ratio once the equilibrium condition of the $1/r$ potential is combined with Landauer’s principle.

2.2 Step 1: Heat released at the boundary (First Law)

The transition from unbound to bound is irreversible. Gravitational collapse is a one-way process; a virialized system does not spontaneously return to the unbound state under the same potential. By the first law of thermodynamics, the heat released irreversibly to the environment during the transition is:

$$Q = E_i - E_f = -E_f = -(\langle K \rangle + \langle V \rangle). \quad (4)$$

This is exact. No approximation has been made. $Q > 0$ requires $E_f < 0$, which is satisfied for any stable bound state.

2.3 Step 2: Entropy produced at the boundary (Second Law)

The transition is irreversible. By the second law of thermodynamics, it produces entropy:

$$\Delta S = \frac{Q}{T_{\text{vir}}}, \quad (5)$$

where T_{vir} is the virial temperature of the bound system. For a self-gravitating system of N particles with velocity dispersion σ , the virial temperature is $T_{\text{vir}} = m\sigma^2/3k_B$ from equipartition, where m is the characteristic particle mass.

2.4 Step 3: Landauer cost of the entropy production

By Landauer's principle [Landauer, 1961], the minimum thermodynamic cost of producing entropy ΔS irreversibly is:

$$E_{\text{Landauer}} = T_{\text{vir}} \Delta S. \quad (6)$$

Substituting Equation 5:

$$E_{\text{Landauer}} = T_{\text{vir}} \times \frac{Q}{T_{\text{vir}}} = Q. \quad (7)$$

The virial temperature cancels exactly. The Landauer cost equals the heat released at the boundary, independently of temperature. This is the first key result: the thermodynamic cost of the transition is exactly Q , regardless of the scale or temperature of the system.

2.5 Step 4: The equilibrium condition of the $1/r$ potential

Steps 1–3 establish that the Landauer cost of the virialization transition equals the heat released at the boundary:

$$E_{\text{Landauer}} = Q = -E_f = -(\langle K \rangle + \langle V \rangle). \quad (8)$$

This is a real and nontrivial result: the thermodynamic cost of the transition from the unbound to the bound state equals the binding energy of the resulting system.

To complete the identification of $\langle K \rangle$, we invoke the equilibrium condition of the $1/r$ potential. For a system governed by a potential $V \propto r^{-1}$, Euler's theorem for homogeneous functions of degree -1 requires:

$$2\langle K \rangle + \langle V \rangle = 0 \implies -E_f = -(\langle K \rangle + \langle V \rangle) = \langle K \rangle. \quad (9)$$

This is the virial theorem stated as the equilibrium condition of the $1/r$ potential — not assumed as a result, but recognized as the condition that defines what it means for a $1/r$ system to be at stable bound equilibrium. Any system not satisfying Equation 9 is either not yet at equilibrium or is unstable.

2.6 Closing the argument

Substituting Equation 9 into Equation 8:

$$\boxed{E_{\text{Landauer}} = Q = -E_f = \langle K \rangle = \frac{1}{2}|\langle V \rangle|} \quad (10)$$

The Landauer cost of the virialization transition equals the heat released at the boundary, which equals the binding energy of the system, which equals the kinetic energy at equilibrium, which equals half the magnitude of the potential energy. The chain is exact at every step.

This is not a derivation of the virial theorem from thermodynamics alone. It is the demonstration that the virial theorem and Landauer’s principle are two expressions of the same identity: the virial theorem provides the mechanical equilibrium condition ($-E_f = \langle K \rangle$); Landauer’s principle provides the thermodynamic identification of what that energy physically is ($E_{\text{Landauer}} = Q = -E_f$). Together they establish that $\langle K \rangle$ is not merely a mechanical quantity but the irreversible thermodynamic cost of the system’s transition to the bound state. The 1/2 partition is the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied for a $1/r$ potential. Any other partition would be inconsistent with either the virial equilibrium condition or the Landauer cost of the transition.

2.7 The physical meaning of K

Equation 10 identifies $\langle K \rangle$ for the first time as a thermodynamic quantity. The kinetic energy of a virialized system is not merely the average kinetic energy of its constituent particles. It is equal in magnitude to the heat Q released irreversibly at the virialization boundary. Q is encoded as entropy on the nearest available surface at Landauer cost $k_B T \ln 2$ per bit. $\langle K \rangle$ is retained in the system as the kinetic energy of the bound particles. Both are exactly half of $|\langle V \rangle|$ — one half leaving the system as irreversible entropy, one half remaining as mechanical motion. These are not two separate processes. They are the two faces of the same partition of $|\langle V \rangle|$ at the virialization boundary.

3 The Thermodynamic Identity: Formal Statement

3.1 The identity

As a system crosses from an unbound to a bound state, it immediately encounters a $1/r$ potential and releases heat Q irreversibly. This heat equals the kinetic energy at equilibrium, $Q = \langle K \rangle$, and Q is encoded onto the nearest available surface as entropy at thermodynamic cost $k_B T \ln 2$ per bit, while $\langle K \rangle$ is retained as the kinetic energy of the bound state. The virial theorem $2\langle K \rangle + \langle V \rangle = 0$ is the mechanical expression of this condition. A stable bound state exists if and only if both are satisfied simultaneously:

$$\langle K \rangle = Q = T \Delta S = E_{\text{Landauer}}. \quad (11)$$

3.2 Comparison to the virial theorem

The virial theorem and the thermodynamic identity are compared in Table 1.

Table 1: Comparison of the virial theorem and the thermodynamic identity.

	Virial Theorem	Thermodynamic Identity
Statement	For a stable bound system, $2\langle K \rangle + \langle V \rangle = 0$	As a system crosses from unbound to bound, $Q = \langle K \rangle = T\Delta S = E_{\text{Landauer}}$; a stable bound state exists if and only if both mechanical and thermodynamic conditions are satisfied simultaneously
Derivation	From Newton's second law (Clausius, 1870)	From the first law, second law, and Landauer's principle (this work)
What it describes	The equilibrium state	The condition for the equilibrium state to exist
Physical content of K	Average kinetic energy of bound particles	Heat released irreversibly at the virialization boundary
Explains the 1/2?	No	Yes: the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied
Scale restriction	None stated; verified mechanically	None: inherited from the scale-invariance of Landauer's principle
Predicts non-virialization?	No: describes only existing bound states	Yes: systems that cannot encode Q on an available surface cannot maintain bound equilibrium

3.3 Why the 1/2 is no longer unexplained

The ratio $\langle K \rangle / |\langle V \rangle| = 1/2$ arises because the $1/r$ potential is homogeneous of degree -1 . Euler's theorem for homogeneous functions gives $2\langle K \rangle = |\langle V \rangle|$ mechanically. The thermodynamic identity shows why this mechanical result has a thermodynamic meaning: the system partitions $|\langle V \rangle|$ exactly in half at equilibrium because that is the unique partition at which the Landauer cost of the transition equals the kinetic energy of the resulting bound state. Any other partition would require either net energy input (unstable) or further energy release (not yet at equilibrium).

3.4 Scale invariance

The scale-invariance of the thermodynamic identity follows directly from the scale-invariance of Landauer's principle. Landauer's principle applies wherever an irreversible process produces entropy, at any scale, at any temperature. The virialization boundary exists at every scale where a $1/r$ potential governs a bound system. Therefore the thermodynamic identity holds at every such scale. This is not an assumption — it is a consequence of the universality of the second law of thermodynamics and Landauer's principle.

4 Evidence Across 37 Orders of Magnitude

Tables 2 and 3 present the evidence for the thermodynamic identity across every physical domain in which bound systems under $1/r$ potentials have been tested. The range extends from hydrogen atoms at 10^{-10} m to the cosmological coupling constant at 10^{26} m — 37 orders of magnitude in physical scale.

Table 2: Core virial theorem domain: the 1/2 partition verified across 37 orders of magnitude. **Blue row**: consistent with 17 independent Planck 2018 MCMC chains with zero free parameters.

System	Scale / Force	$K/ V $	Result	Precision
Electron rest mass	λ_C , EM	Fixed point	m_e to 6.6 ppm	Zero free parameters [Mahaffey, 2026b]
Atoms (H–Xe, 20 elements)	10^{-10} m, EM	$T/ V $	1.0000000000	Machine precision [Clementi and Roetti, 1974, Bunge et al., 1993]
Molecules (H ₂ –CO ₂ , 10)	10^{-9} m, EM	$T/ V $	1.0000000000	Exact at equil. geom. [Clementi and Roetti, 1974, Bunge et al., 1993]
Equipartition theorem	10^{-9} – 10^{-2} Thermal	m, $k_B T/2$ per DOF	1/2 exact	Exact in limit (harmonic potential; same ratio, different theorem)
Solar structure	10^9 m, Gravity	$K/ U $	≈ 0.5	$\sim 10\%$ [Mahaffey, 2026g]
White dwarfs (Chandrasekhar)	10^7 m, Gravity	M_{Ch}	$1.44 M_\odot$	Exact, observed [Chandrasekhar, 1931]
Galaxy clusters (~ 20)	10^{23} m, Gravity	$K/ U $	≈ 0.5	$\sim 20\%$ [Mahaffey, 2026g]
Virial efficiency $\langle \eta \rangle$ (6 N-body)	10^{22} – 10^{25} Gravity	m, $2K/ U $	0.815 ± 0.025	6 independent sims [Bryan and Norman, 1998, Bett et al., 2007, Neto et al., 2007, Ludlow et al., 2010, Power et al., 2012, Klypin et al., 2016]
Cosmological coupling β_m	10^{26} m, Gravity	β_m/Ω_m	0.1583 ± 0.0033	0.2 σ , 17 chains [Mahaffey, 2026d]
Scale range	λ_C (electron) to 10^{26} m (cosmic horizon) — >37 orders of magnitude			

Table 3: Extended domain: expressions of the thermodynamic identity beyond the classical virial theorem. **Green rows**: results derived within the IAM framework. **Red row**: structural exception (see Section 5).

Domain	Scale	Expression	Result
H_0 sector split	Cosmological	$H_0^{\text{matter}}/H_0^{\text{photon}}/\sqrt{1+\beta_m}$	$= 72.26/67.16$ $\text{km s}^{-1} \text{Mpc}^{-1}$, both $< 1\sigma$ [Mahaffey, 2026d]
σ_8 suppression	Cosmological	$\sigma_8^{\text{IAM}} = 0.800$	0.7998 ± 0.0058 , 0.1σ [Mahaffey, 2026d]
M- σ relation (SMBHs)	10^{20} m	$M_{\text{BH}} \propto \sigma^4$	2% from observed, 0.7σ (115 galaxies) [Mahaffey, 2026c]
BH entropy partition	$r_s = 2GM/c^2$	$E_{\text{Landauer}} = (\ln 2/2)M_{\text{BH}}c^2$	34.7% of rest mass (universal) [Mahaffey, 2026a]
Dark sector virial partners	$a = 1$	$\Omega_m/[\beta_m\mathcal{E}(1)] = 2$ exactly	Exact, zero free parameters [Mahaffey, 2026h]
Weak nuclear force	10^{-18} m	Structural exception: cannot satisfy equilibrium condition	Consistent with identity’s physical content (Sec. 5)

5 The Four Fundamental Forces

The thermodynamic identity is satisfied by every force that maintains stable bound structures. Three of the four fundamental forces fall into this category.

Gravity operates through a $1/r$ potential in three-dimensional space. Gauss’s law requires the $1/r^2$ force from any point source in 3D, giving the $1/r$ potential and therefore the exact virial ratio $1/2$ for any gravitationally bound system. The thermodynamic identity is verified from stellar structure through galaxy clusters to the cosmological coupling constant at 0.2σ from 17 independent MCMC chains.

Electromagnetism operates through the same $1/r$ Coulomb potential. The thermodynamic identity is verified at machine precision for 20 atoms and 10 molecules from published Hartree–Fock energies [Clementi and Roetti, 1974, Bunge et al., 1993]. The mean virial ratio $\eta = T/|E| = 1.0000000000$ with standard deviation zero. For converged Hartree–Fock wavefunctions, the virial ratio must equal unity exactly as a mathematical consequence of the $1/r$ potential — an exact verification of the thermodynamic identity at the quantum level.

The strong nuclear force produces the first virialized structures in cosmic history: protons and neutrons forming at $t \approx 10^{-6} \text{ s}$ when the quark-gluon plasma underwent QCD confinement. The resulting nucleons are bound states satisfying the thermodynamic identity within the strong-force domain. Note that the Chandrasekhar white dwarf limit — though commonly cited in this context — arises from the equilibrium between electron degeneracy pressure and gravitational collapse, and is therefore a gravitational verification of the thermodynamic identity, not a strong-force one. It is listed in Table 2 under gravity.

The weak nuclear force is the structural exception. It violates parity [Wu et al., 1957], violates CP symmetry [Christenson et al., 1964], and is responsible for beta

decay, stellar nucleosynthesis, and the matter-antimatter asymmetry that produced the observable universe [Sakharov, 1967]. It cannot satisfy the thermodynamic identity not because it fails to do so empirically but because satisfying it would prevent the weak force from performing its physical function.

The thermodynamic identity describes the equilibrium condition of stable bound states. The weak force’s function is to enable irreversible transitions between stable states — to dissolve one equilibrium and create another. A force that satisfies the equilibrium condition maintains the current state. The weak force by definition transforms the current state. These functions are structurally incompatible.

The exception is evidence, not a counterexample. A thermodynamic identity that holds for all forces governing stable structures and is violated by precisely the one force whose function is structural transformation distinguishes equilibrium forces from transition forces on physical grounds.

6 The MCMC Confirmation

The most rigorous quantitative test of the thermodynamic identity is the Planck 2018 MCMC posterior returning $\beta_m = 0.1583 \pm 0.0033$ against the theoretically predicted $\beta_m = \Omega_m/2 = 0.1577$ — a confirmation at 0.2σ — from 17 independent converged chains with no free parameters, no discarded runs, and no excluded redshift ranges [Mahaffey, 2026d].

The identity predicts $\beta_m = \Omega_m/2$ from the virial partition of gravitational decoherence energy, before any chains were run. The Planck 2018 MCMC sampler has no knowledge of the thermodynamic identity, the virial theorem, or the physical argument that $\beta_m = \Omega_m/2$. It samples the posterior probability distribution of the cosmological parameters given the Planck 2018 likelihood. The posterior peak at $\beta_m = 0.1583$ arrives there because the data are consistent with that value.

This sequence — derivation from first principles before observation, the data consistent with the prediction from an independent measurement with no knowledge of the prediction — constitutes the standard of scientific evidence. The confirmation at 0.2σ across 17 independent chains is the strongest available quantitative test of the thermodynamic identity at cosmological scales.

Table 4: MCMC confirmation of $\beta_m = \Omega_m/2$. All 17 chains converged to $R - 1 < 0.01$. No chains discarded. Zero free parameters beyond Λ CDM.

Chain set	Dataset	β_m posterior vs prediction (0.1577)
Level 1 (12 chains)	Planck 2018 + combinations	0.1577–0.1592 (< 0.3% from prediction)
Level 2 Run A	Planck 2018 full likelihood	0.1583 ± 0.0033 (0.2σ)
Level 2 Run D	Planck 2018 + RSD	0.1583 ± 0.0033 (0.2σ)
All 17 chains	All combinations	Consistent at $< 0.3\sigma$, zero free parameters

The best-fit $\Delta\chi^2 = +0.54$ relative to Λ CDM — with β_m fixed by the thermodynamic identity and zero additional free parameters — indicates that the IAM framework, which

instantiates this identity at cosmological scales, is statistically indistinguishable from Λ CDM under the full Planck likelihood while simultaneously providing a physical account of the σ_8 and H_0 tensions that Λ CDM does not address [Mahaffey, 2026d].

An independent cross-check from N-body simulations confirms the same prediction. Six independent simulation studies [Bryan and Norman, 1998, Bett et al., 2007, Neto et al., 2007, Ludlow et al., 2010, Power et al., 2012, Klypin et al., 2016] spanning 25 years of published literature measure the mass-weighted virial efficiency $\langle\eta_{\text{vir}}\rangle = 0.815 \pm 0.025$, consistent with the thermodynamic identity prediction $\eta_{\text{vir}} = 1/(2f_{\text{coll}}) = 0.81$ using the published collapsed fraction $f_{\text{coll}} = 0.62 \pm 0.03$ [Tinker et al., 2008]. The combined measurement $\Omega_m \times f_{\text{coll}} \times \langle\eta_{\text{vir}}\rangle = 0.315 \times 0.62 \times 0.815 = 0.159 \pm 0.010$ agrees with $\beta_m = \Omega_m/2 = 0.1575$ to 1.0% with no free parameters [Mahaffey, 2026i].

7 Discussion

7.1 The virial theorem as a special case

The derivation in Section 2 shows that the virial theorem is the mechanical expression of a thermodynamic identity. The identity is more general: it governs the boundary between unbound and bound states at every scale where a $1/r$ potential produces stable equilibrium. The virial theorem describes the equilibrium state. The thermodynamic identity describes the condition for that state to exist and to persist.

This distinction has physical consequences. The virial theorem is silent on what happens when the conditions for equilibrium cannot be satisfied. The thermodynamic identity predicts the outcome directly: a system that cannot encode Q on an available surface cannot maintain bound equilibrium and returns to the unbound state. This prediction is consistent with the missing satellites problem, where dark matter halos below a critical velocity dispersion $\sigma_{\text{crit}} \approx 4 \text{ km s}^{-1}$ disperse rather than virialize, because no local black hole horizon is available to absorb the entropy produced at the virialization boundary [Mahaffey, 2026e].

7.2 Landauer’s principle as the physical basis for scale invariance

The virial theorem has been verified across 37 orders of magnitude without a physical explanation for why it should hold across such a range. The thermodynamic identity provides that explanation. Landauer’s principle applies wherever an irreversible process produces entropy. The virialization boundary exists wherever a $1/r$ potential produces a bound state. The boundary condition $\langle K \rangle = Q$ therefore holds at every such scale, from the hydrogen atom to the cosmic horizon, because Landauer’s principle is scale-invariant and the $1/r$ potential form is universal.

This addresses the question of why the virial theorem should apply at cosmological scales. It is not an extension of a mechanical result beyond its domain. It is a thermodynamic identity that applies wherever its conditions are met, at any scale.

7.3 Relationship to Jacobson 1995

Jacobson [1995] derived the Einstein equations from the thermodynamic identity $\delta Q = T \delta S$ applied to the Rindler horizon of an accelerated observer, with $S = A/4\ell_P^2$. That derivation showed that Einstein’s equations are thermodynamic in origin.

The present work applies the same thermodynamic identity $Q = T \Delta S$ to the virialization boundary of a self-gravitating system and establishes that the virial theorem is its mechanical expression. Both start from the same thermodynamic identity. Both show that a fundamental result of classical physics has thermodynamic origin. The present work extends the Jacobson program from the field equations governing curved spacetime to the equilibrium condition governing bound states within that spacetime.

8 Conclusion

The virial theorem $2\langle K \rangle + \langle V \rangle = 0$ has been verified across 37 orders of magnitude for 154 years without a physical identification of what $\langle K \rangle$ represents thermodynamically.

This paper shows that the virial theorem and Landauer’s principle are two expressions of the same thermodynamic identity. The first law establishes $Q = -E_f$; the second law and Landauer’s principle establish $E_{\text{Landauer}} = Q$; the equilibrium condition of the $1/r$ potential establishes $-E_f = \langle K \rangle$. Together these require:

$$\langle K \rangle = Q = T \Delta S = E_{\text{Landauer}}.$$

$\langle K \rangle$ is the heat released irreversibly at the virialization boundary, encoded as entropy on the nearest available surface at Landauer cost $k_B T \ln 2$ per bit. The virial theorem is the mechanical expression of this thermodynamic identity.

The virial theorem is the mechanical expression of this thermodynamic identity. The $1/2$ partition is the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied for a $1/r$ potential. Its appearance across 37 orders of magnitude is a consequence of the scale-invariance of Landauer’s principle and the universality of the $1/r$ potential form.

Three of the four fundamental forces satisfy the identity. The weak nuclear force is the structural exception, excluded by physical necessity. The data are consistent with the identity at 0.2σ at cosmological scales from 17 independent MCMC chains with zero free parameters.

We propose that this thermodynamic identity constitutes a fundamental law of physics of which the virial theorem is the mechanical special case.

Acknowledgments

The thermodynamic framework of T. Jacobson [Jacobson, 1995] is the foundation upon which this work is built. The author thanks R. Landauer [Landauer, 1961] and R. Clausius [Clausius, 1870] for the tools that make this connection possible.

Data Availability

All MCMC chains, analysis scripts, and supporting papers are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation> with canonical Zenodo DOI [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042).

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